

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Matrices

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2 . (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **ICSE 2025, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Answer: (c)

2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. **ICSE 2025, 4 marks**

SOLUTION

$$(a) A + C = \begin{bmatrix} 1-5 & 2+1 \\ 3+7 & 4-4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$$

$$(b) B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix} = \begin{bmatrix} -8+10 & 6+0 \\ -16+20 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$$

$$(c) 5B = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$$

$$(d) B(A + C) - 5B = \begin{bmatrix} 2-10 & 6-5 \\ 4-20 & 12-10 \end{bmatrix}$$

$$\text{Answer: (d)} \begin{bmatrix} -8 & 1 \\ -16 & 2 \end{bmatrix}$$

3. If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, find x . (a) 2 (b) 4 (c) 8 (d) 10 **ICSE 2024, 1 mark**

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4+0 & 4+4 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 0 & 4 \end{bmatrix}$$

Answer: (c)

4. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ 1 & 5 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find $A(B + C) - 14I$. **ICSE 2023, 3 marks**

SOLUTION

$$B + C = \begin{bmatrix} 5 & 3 \\ 3 & 9 \end{bmatrix}$$

$$A(B + C) = \begin{bmatrix} 5 + 9 & 3 + 27 \\ 10 + 12 & 6 + 36 \end{bmatrix} = \begin{bmatrix} 14 & 30 \\ 22 & 42 \end{bmatrix}$$

$$14I = \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} 0 & 30 \\ 22 & 28 \end{bmatrix}$$

5. If $A = \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 2 \\ 1 & 0 \end{bmatrix}$, find $A^2 - 2AB + B^2$. **ICSE 2020, 3 marks**

SOLUTION

$$A^2 = \begin{bmatrix} 9 + 0 & 0 + 0 \\ 15 + 5 & 0 + 1 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 20 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -12 + 0 & 6 + 0 \\ -20 + 1 & 10 + 0 \end{bmatrix} = \begin{bmatrix} -12 & 6 \\ -19 & 10 \end{bmatrix}, \text{ so } 2AB = \begin{bmatrix} -24 & 12 \\ -38 & 20 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 16 + 2 & -8 + 0 \\ -4 + 0 & 2 + 0 \end{bmatrix} = \begin{bmatrix} 18 & -8 \\ -4 & 2 \end{bmatrix}$$

$$A^2 - 2AB + B^2 = \begin{bmatrix} 9 + 24 + 18 & 0 - 12 - 8 \\ 20 + 38 - 4 & 1 - 20 + 2 \end{bmatrix}$$

$$\text{Answer: } \begin{bmatrix} 51 & -20 \\ 54 & -17 \end{bmatrix}$$

2 Solving matrix equations

6. $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$. Find x and y if $AB = C$. **ICSE 2024, 4 marks**

SOLUTION

$$AB = \begin{bmatrix} 4x + 0 & 0 + 0 \\ 4 + y & 0 + 1 \end{bmatrix} = \begin{bmatrix} 4x & 0 \\ 4 + y & 1 \end{bmatrix}$$

Compare with C : $4x = 4$, so $x = 1$. Then $4 + y = x = 1$, so $y = -3$.

Answer: $x = 1, y = -3$

7. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, find x and y . (a) 1, -2 (b) -2, 1 (c) 1, 2 (d) -2, -1 **ICSE 2023, 1 mark**

SOLUTION

$$\text{Multiply out: } \begin{bmatrix} 2x \\ 4y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}, \text{ so } x = 1 \text{ and } y = -2.$$

Answer: (a)

8. $A = \begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ and $A^2 = 3I$, where I is the identity matrix of order 2. Find x and y . **ICSE 2020, 3 marks**

SOLUTION

$$A^2 = \begin{bmatrix} x^2 + 3y & 3x + 9 \\ xy + 3y & 3y + 9 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$3x + 9 = 0$ gives $x = -3$, and $3y + 9 = 3$ gives $y = -2$.

Check: $x^2 + 3y = 9 - 6 = 3$ and $xy + 3y = 6 - 6 = 0$.

Answer: $x = -3, y = -2$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.